

AI FDE Operating Model ..

Seq.	Workflow	Methods and disciplines	Essential artifacts	Output/input to next phase
1	Mandate and field immersion	FDE ownership · Stakeholder discovery · Responsible AI context · ISO/IEC 42001 scope	Engagement charter · Scope · Outcome statement · Sponsor/owner · Governance RACI · Stakeholder and affected-groups maps · Field-evidence register	Approved mandate and operating context
2	Discover process and architecture	Process Discovery · Lean · Waste Elimination · DMAIC Define/Measure · Architecture Discovery · Brownfield repo assessment · C4	SIPOC · Process/value-stream maps · Waste register · System landscape · Brownfield assessment · Current-state C4 views · Dependencies · Data flows · Trust boundaries	Current-state process and architecture baseline
3	Frame problem, root cause and value	SCQA · DMAIC Analyse · Root-cause analysis · Baseline KPIs · Data-driven framing	SCQA problem frame · Root-cause analysis · Baseline dataset · KPI tree · CTQs · Value hypothesis · Counter-metrics · Success/failure criteria	Evidence-backed problem and baseline
4	Triage regulation and qualify use case	Responsible AI · ISO/IEC 42001 · EU AI Act classification · AI suitability · Solution Decoding	Impact/regulatory screen · Prohibited-use check · AI suitability assessment · Non-AI alternative · Use-case card · Value-risk-feasibility matrix · Go/no-go and kill criteria	Approved and justified use case
5	Model the domain	Domain-Driven Design · Ubiquitous Language · Bounded Contexts · Domain Events	Glossary · Domain capability map · Business rules · Decision model · Domain events · Ownership map · DDD Context Map · Bounded Contexts	Domain and decision model
6	Qualify data and knowledge	Data Discovery · Quality · Lineage · Provenance · Permissible use · Knowledge discovery	Data/knowledge inventories · Lineage · Quality profile · Provenance · Access matrix · Representativeness assessment · Data-gap register · Dataset datasheets	Data and knowledge readiness assessment
7	Define evaluations, impacts and risks	Evals · Responsible AI · ISO/IEC 42005 · ISO/IEC 42001 risk planning · EU AI Act requirements	Evaluation strategy · Golden-set specification · Scenarios · Acceptance thresholds · AI impact assessment · Risk/harms registers · Risk treatment · Oversight and transparency requirements	Evaluation, impact and risk requirements

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8	Generate, test and select options	Solution Decoding · Reference Architectures · GenAI patterns · PoCs · Trade-off Matrices	Solution catalogue · Reference-architecture comparison · PoC/ model/RAG results · Weighted trade-off matrix · Build/buy assessment · Provider comparison · Preliminary ADRs · Selected solution	Approved solution and trade-offs
9	Design information architecture	Semantic Layers · Ontology · Knowledge Graphs · Graph/Vector Databases · Data Architecture Decisions	Target data architecture · Data contracts · Semantic model · Metadata/ provenance design · Retrieval architecture · Data ADRs · Conditional ontology and graph/vector schemas	Approved information architecture
10	Design AI and application architecture	C4 · GenAI patterns · RAG · Model routing · Prompt/context design · Integration/ deployment patterns - artifct x from 42001	Target C4 Context/ Container/Component views · AI/RAG architecture · Model-routing design · API contracts · Prompt/ context design · Deployment topology · Failure-mode design · Architecture ADRs	Complete base AI/ application architecture
11	Design agentic and multi-agent orchestration	Agentic AI · Single-agent versus multi-agent decision · Multi-Agent Orchestration · Human-in-the-loop · Agent communication and control	Agent-suitability assessment · Autonomy-level ADR · Agent responsibility map · Orchestration topology · Agent interaction/sequence diagram · State-machine model · Tool/ action catalogue · Identity/permission matrix · Shared-memory design · Handoff protocol · Termination/loop controls · Human approval/override/ escalation matrix	Approved, bounded and testable agentic architecture
12	Design security, guardrails and supplier controls	AI Security · Threat Modelling · OWASP GenAI LLM Top 10 2026 · OWASP Agentic Top 10 2026 · Guardrails · Supply-chain assurance	Threat model · Agentic attack-surface map · Trust boundaries · Abuse cases · OWASP mapping · Guardrail architecture · Control matrix · Supplier/ model assessment · Model/system cards · Component register · SBOM/AIBOM · Exit plan	Controlled architecture and approved components
13	Approve ADRs and delivery specification	ADRs · Spec-Driven Development · Telemetry/	Final ADR register · Approved C4 baseline · Delivery specification · NFRs · API/data	Build-ready specification

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		Observability design · Traceability	contracts · Agent specifications · Evaluation cases · Telemetry/logging specification · Rollback requirements · Traceability matrix · Backlog	
14	Engineer	<u>AI-Assisted Engineering · Spec-Driven Development using Cursor · Brownfield integration · Agent/workflow implementation</u>	Source/laC · Prompt/configuration registry · Model/tool versions · Data pipelines · Agent definitions · Orchestration workflows · Unit/integration tests · CI/CD evidence · As-built C4 views	Versioned deployable increments
15	Evaluate, attack and independently assure	Model/RAG/agent/multi-agent Evals · Responsible AI testing · OWASP testing · Red teaming · Guardrail validation	Evaluation harness · Golden datasets · Model/RAG results · Agent task-completion results · Inter-agent handoff tests · Coordination/conflict tests · Loop/termination tests · Prompt-injection/tool-misuse tests · Assurance report · Residual-risk acceptance	Independently assured release candidate
16	Prepare operations, recovery and regulatory evidence	Telemetry · Observability · Change management · Pre-production Chaos Drill · ISO/IEC 42001/42005 · EU AI Act	Operational RACI · Agent/LLM telemetry · Dashboards/alerts · Runbooks · Incident/rollback plans · SOPs/training · Recovery evidence · AI-system record · Technical documentation · Transparency and oversight records	Operationally and legally ready release
17	Deploy progressively and integrate adoption	Shadow/pilot/canary · Bounded agent autonomy · Human–AI workflow integration	Release manifest · Deployment record · Pilot/canary results · Autonomy-expansion record · Rollback decision · Adoption dashboard · Override/workaround logs · Updated SOPs	Controlled live service and adoption evidence
18	Monitor and validate operational resilience	Telemetry · Observability · Continuous Evals · Agent monitoring · Drift · Controlled Chaos Drills	Operational dashboards · Model/tool/agent traces · Inter-agent interaction logs · Drift reports · Cost/availability reports · Incident records · Agent-loop alerts · Chaos/recovery results · Updated runbooks	Operational performance, risk and resilience evidence

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19	Prove value and tell the decision story	KPIs before/after · DMAIC Improve · Lean · SCQA · Data-Driven Storytelling	Benefits-realisation report · Before/after KPIs · Cost-to-value analysis · Counter-metrics · Waste reduction · Unintended effects · Executive SCQA paper · C4 executive view · Decision recommendation	Evidence-backed scale/change/stop package
20	Evaluate AIMS and decide lifecycle state	ISO/IEC 42001 Check/Act · Internal audit · Management review · DMAIC Control · Change reassessment	AIMS performance report · Audit findings · CAPA register · Management-review decision · Control plan · Updated impact/risk/C4/ADRs/evaluations · Scale/change/restrict/suspend/retire decision	Authorised lifecycle state or return to an earlier phase
21	Retire and capture reusable IP	Safe retirement · Record control · Lessons learned · Reference Architecture improvement	Retirement plan · Notifications · Access revocation · Data/model/memory disposition · Agent/tool credential revocation · Supplier/dependency closure · Reusable ADRs/C4 templates · Updated reference architectures · Eval and guardrail templates	Formally closed system and reusable FDE IP

FDE - Releasing to the customer - 6 months repo 3.0